

SMI 2026 Program Book

Track A · S1 | 8:00–9:20 AM, June 1 | Michigan Room, 2nd floor

Scalable, Interpretable, and Reproducible Methods for Neuroimaging Analysis

Yuliang Xu	<i>Bayesian Image Regression with Soft-thresholded Conditional Autoregressive Prior</i>
Luling Altan	<i>Alzheimer's Disease Subtype Identification from Spatial Tau-PET Heterogeneity Using Self-Supervised Deep Learning</i>
Emily Hector	<i>A new block covariance regression model and inferential framework for massively large neuroimaging data</i>
Paul Taylor	<i>Go Figure: Transparency in neuroscience images preserves context and clarifies interpretation</i>

Track B · S2 | 8:00–9:20 AM, June 1 | Hussey Room, 2nd floor

Advances in methods for cancer imaging: From single-cell data to sustainable AI

Inna Chervoneva	<i>Spatial index biomarkers based on quantitative single-cell protein expression imaging data</i>
Lubomir Hadjiski	<i>Integrated approach: Large language models, deep learning and radiomics for survival prediction of bladder cancer patients</i>
Andrew Whiteman	<i>Using computer vision-inspired kernels to predict cancer prognosis from multiplexed images of tumor micro-environments</i>
Shu (Joy) Jiang	<i>Sustainable AI in cancer imaging: From research to market</i>

Track A · S3 | 9:30–10:50 AM, June 1 | Michigan Room, 2nd floor

AI for Medical Image Analysis I

Xiaoxing Ren	<i>A Communication-Efficient Decentralized Learning Algorithm for Multi-Center Medical Image Classification</i>
John Reavey-Cantwell & Emily Wittrup	<i>An Integrative AI Method for Stroke Treatment Assessment and Management</i>
Amoon Jamzad	<i>Multimodal Intraoperative Margin Assessment in Breast Conserving Surgery</i>
Robert Policelli	<i>A pathway-based approach to improve breast cancer classification explainability</i>

Track B · S4 | 9:30–10:50 AM, June 1 | Hussey Room, 2nd floor

Statistical Innovation in Multiplex Imaging: Methods for Robust Discovery

Thao Vu	<i>Spatial and functional analytic approaches for single-cell imaging data using entropy measures</i>
Souvik Seal	<i>SpaceBF: Spatial coexpression analysis using Bayesian Fused approaches in spatial omics datasets</i>
Siyuan Ma	<i>Batch-Agnostic Self-Supervised Learning Uncovers Clinically Relevant Tumor Niches in Multiplexed Imaging</i>
Ruoqian Liu	<i>Detecting Disease-Associated Dysbiosis in Image-Based Spatial Microbiome Data</i>

Track C · S5 | 9:30–10:50 AM, June 1 | Room 4, 1st floor

Alignment, Calibration, and Robust Inference in Neuroimaging

Zhengwu Zhang	<i>Alignment of Continuous Brain Connectivity</i>
Yiwen Dong	<i>Alzheimer's Disease Biomarkers Harmonization Challenges</i>
Andrew Chen	<i>Calibration of MRI-Based Reference Intervals to New Samples</i>
Xinyu Zhang	<i>Asymptotic Distribution of Robust Effect Size Index</i>

11:15 AM–12:15 PM, June 1 | Michigan League Ballroom, 2nd floor

Keynote Presentation #1

Hernando Ombao	<i>Spectral Measures of Brain Functional Connectivity</i>
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Track A · S6 | 1:30–2:50 PM, June 1 | Michigan Room, 2nd floor

Spatiotemporal Models for Brain Structure, Function, and Connectivity

Mara Sherlin Talento	<i>KenCoh: A Rank-Based Estimator of Coherence between Multivariate Time Series</i>
Yuhyeong Jang	<i>A Survival Analysis of Glioma Patients using Topological Features</i>
Heather Shappell	<i>A Hidden Semi-Markov Model Framework for Covariate-Dependent Sojourn Distributions in Dynamic Brain Network Analysis</i>
Joshua Lukemire	<i>Hierarchical Modeling of Localized Intracranial Volume Abnormalities in Craniosynostosis</i>

Track B · S7 | 1:30–2:50 PM, June 1 | Hussey Room, 2nd floor

Advanced Statistical and Machine Learning based methods for Spatial imaging and omics

Satwik Acharyya	<i>Network models for Spatial omics data</i>
Ronglai Shen	<i>Spatial Immunophenotyping from Whole-Slide Multiplexed Tissue Imaging using Convolutional Neural Networks</i>
Michele Peruzzi	<i>Integrated Geospatial Network models for Spatial Omics</i>
Veera Baladandayuthapani	<i>Bayesian Statistical Models for Spatial Imaging in Cancer</i>

Track C · S8 | 1:30–2:50 PM, June 1 | Room 4, 1st floor

Principled Statistical and Machine Learning Methods for Brain Imaging Inference

Benjamin Risk	<i>Causal inference for hierarchical machine learning with an application to stimulant impacts on brain connectivity in autism</i>
Armin Iraj	<i>From Signals to Sources: Advances in Dynamic Precision Functional Mapping</i>
Guorong Wu	<i>Uncover Governing Law of Pathology Propagation Mechanism Through A Mean-Field Game</i>
Will Consagra	<i>Learned Hemodynamic Coupling Inference in Resting-State Functional MRI</i>

Track A · S9 | 3:00–4:20 PM, June 1 | Michigan Room, 2nd floor

Advanced statistical and AI models for complex and graphical brain images

Yuexuan Wu	<i>High-Dimensional Multivariate Mediation Analysis for Brain Imaging via Structured Dimension Reduction</i>
Zhihao Wu	<i>Dependence-Aware Knockoff Enables Stable Feature Discovery in Brain Imaging</i>
Qiong Wu	<i>Distributed Covariance Graph-Guided ComBat (dG-ComBat): A Privacy-Preserving Framework for Harmonizing Multi-Site Neuroimaging Data</i>
Xinyuan Tian	<i>Estimating a shared backbone linking structural and functional connectomes via bidirectional graph autoencoders</i>

Track B · S10 | 3:00–4:20 PM, June 1 | Hussey Room, 2nd floor

Emerging Statistical Techniques for Longitudinal and Multimodal Imaging Analysis

Ivo Dinov	<i>Kime-Phase Tomography and Manifold Representation of Longitudinal Processes over a Complex-Time Domain</i>
Rajarshi Guhaniyogi	<i>DeepSpFactor: A Statistically Trustworthy Deep Learning Framework for Spatial Factor Models in Multi-subject Multi-modal Imaging</i>
Simon Vandekar	<i>Semiparametric confidence sets for a robust effect size in Neuroimaging</i>
Sharmistha Guha	<i>Multimodal Neuroimaging in Aging: Joint Learning of Network Data and Spatially Correlated Node Attributes</i>

Track C · S11 | 3:00–4:20 PM, June 1 | Room 4, 1st floor

Modern Statistical Methods for Neuroimaging

John Kornak	<i>An MCMC Approach to Bayesian Image Analysis in Fourier Space</i>
Moo Kyung Chung	<i>Causality without DAG</i>
Dayu Sun	<i>Sparse Partial Generalized Tensor Regression With Application to Neuroimaging Data</i>
Gang Chen	<i>Navigating Correlation, Causation, and Interpretation in Resting-State fMRI</i>

4:30–6:30 PM, June 1 | Various Locations

Training Tutorials & Short Courses

Empowering Large Language Models with Statistics: A Practical Tutorial for Medical Imaging

Khashayar Namdar & Pascal Tyrrell | *Michigan League Ballroom, 2nd floor*

Manifold learning and dimension reduction for imaging data

Chunming Zhang | *Hussey Room, 2nd floor*

Methods for FMRI processing, quality control and group analysis in AFNI

Paul Taylor & Gang Chen | *Michigan Room, 2nd floor*

Introduction to P300 Brain-Computer Interface Data Analysis Challenges

Jane Huggins | *Room 4, 1st floor*

Track A · S12 | 8:00–9:20 AM, June 2 | Michigan Room, 2nd floor

Methods for Statistical Imaging Data with Applications to Physical Sciences

David van Dyk	<i>Identification of high-energy astrophysical point sources via hierarchical Bayesian nonparametric clustering</i>
Hengrui Luo	<i>Bayesian Partition-Based Scalar-on-Image Regression with Group Spike-and-Slab Priors</i>
Jeffrey Regier	<i>Expectation-Maximization with Dual-Stream Flow Matching for Cataloging Astronomical Images</i>
Rachel Mandelbaum	<i>Precise estimation of weak gravitational lensing distortions from images</i>

Track B · S13 | 8:00–9:20 AM, June 2 | Hussey Room, 2nd floor

From Connections to Understanding: Brain Network Models in Neuroscience

Yi Zhao	<i>Estimation of Heterogeneous Causal Mediation Effects</i>
Yize Zhao	<i>Bayesian graph-informed modeling for Tau-connectivity interaction</i>
Mohsen Bahrami	<i>Brain Networks: Multivariate Statistical Tools to Study Structure, Function, and Dynamics</i>
Noah Gade	<i>Segmentation in Dynamic Neuroimaging via Random Featurizations</i>

Track C · S14 | 8:00–9:20 AM, June 2 | Room 4, 1st floor

Advanced Statistical Modeling and Machine Learning for Brain Imaging Data

Aiying Zhang	<i>ConnectMVR: A Supervised Brain Connectivity Analysis Framework for Predicting Behavioral Outcomes</i>
Shuo Chen	<i>Trustworthy ML/AI for Aging Clocks: Preventing Systematic Prediction Bias in Biological Age Estimation</i>
Yuan Wang	<i>Topological Inference and Clustering of Functional Brain Networks</i>
Ruiwen Zhou	<i>A Temporal Graph Neural Network for Time-to-Alzheimer's Disease Prediction Using Longitudinal Neuroimaging Data</i>

Statistical Method for Interpreting Astronomical Images

Kevin McKinnon	<i>Measuring precise motions of the faintest stars by combining Hubble, James Webb, and Gaia</i>
Tamás Budavári	<i>Towards Cleaner and Deeper Sky Reconstructions from Astronomy Exposures: ImageMM and Beyond</i>
Andrew Saydjari	<i>Separating Stars and the Diffuse Sky: Leveraging Thermal Light Echoes for 3D Tomography</i>
Yang Chen	<i>Probabilistic Tensor Prediction for High-Dimensional Solar Imaging Data</i>

Statistical and AI Methods for Heterogeneous and Irregular Imaging Data

Haochang Shou	<i>Adaptive Shrinkage Estimation for Personalized Deep Kernel Regression in Modeling Brain Trajectories</i>
Liyue Shen	<i>Constrained and Controllable Diffusion Models for Computational Imaging</i>
Christopher Geoga	<i>Sharpening and accelerating Fourier methods for irregularly sampled measurements</i>
John Bodenschatz	<i>Mathematically Accurate Maximum Likelihood Estimation Detects Activation in fMRI Phase Data from Local Magnetic Field Changes</i>

Achieving Reliable Inference and Prediction from Noisy Imaging Data through Structure Learning

Fangyi Wang	<i>Joint Registration and Conformal Prediction for Partially Observed Functional Data</i>
Megan Jones	<i>Estimation of Maximum Achievable Predictive Accuracy for Machine Learning Brain-Phenotype Associations</i>
Yezhi Pan	<i>Network-Guided Multivariate Regression for High-dimensional Imaging Outcomes</i>
Jiaao Yu	<i>Structured multi-modal imaging interaction discovery with a triple-graph mode</i>

Keynote Presentation #2

Nicole Lazar

On the Replicability of Functional Brain Networks

Track A · S18 | 1:30–2:50 PM, June 2 | Michigan Room, 2nd floor

Recent Advances in Complex Modeling of Brain Imaging and Networks

Panpan Zhang

Group-Structured Sparse Precision Estimation for Interpretable Functional Brain Networks

Marina Vannucci

Bayesian Brain Network Mediation Analysis of Longitudinal Alzheimer's Disease Data

Suprateek Kundu

Bayesian Semi-parametric Tensor-on-Tensor Regression for Forecasting Alzheimer's Disease Progression

Yu-Ping Wang

Nonlinear mixed models for brain network analysis based on Multi-Paradigm fMRIs

Track B · S19 | 1:30–2:50 PM, June 2 | Hussey Room, 2nd floor

Next-Generation Statistical and Inferential Methods for Neuroimaging and Multimodal Data Integration

Zhirui Li &
Bingxin Zhao

Agentic AI in imaging genetics

Jeffery Morris

Scalable distributional regression

Quy Cao

Spatial Bootstrap for Voxel-Wise Neuroimaging Analysis

Noah Hillman

Testing for Network Specificity in Brain-Behavior Associations Using Ordinal Dominance Curves

AI for Medical Image Analysis II

Morgen Henry	<i>A Four-Stage Statistical Pipeline for Medical Image Super-Resolution via Generative Adversarial Learning</i>
Khashayar Namdar	<i>RadiomiXAI: adjusting radiomics feature importance via Bayesian Network-guided explainability</i>
Amoon Jamzad	<i>Spatial Prompting for Representation and Exploration of High-Dimensional Data</i>

Recent advancements in statistical methods for MRI

Seonjoo Lee	<i>Model Selection for Exposure-Mediator Interaction for high-dimensional imaging mediators</i>
Chao Huang	<i>Dynamic Mediation Analysis</i>
Kaidi Kang	<i>Characterizing Longitudinal Trajectories of Structural MRI-derived Biomarkers Toward Alzheimer's Disease Progression Using a Double anchoring events-based Sigmoidal Mixed Model</i>
Jun Young Park	<i>Statistical considerations in intermodal coupling analysis</i>

Statistical Network Analysis for Neuroimaging

Liza Levina	<i>Comparing groups of networks</i>
Jaroslav Harezlak	<i>Simultaneous Nuclear Norm and L_1 Regularization for Binary-outcome-on-Matrix Regression with Applications to Brain Imaging</i>
Keith Levin	<i>Estimating Multiple Weighted Networks with Node-Sparse Differences and Shared Low-Rank Structure</i>
Braden Scherting	<i>Quasi-Poisson multiplex ANOVA with applications to brain connectors</i>

Recent developments on neuroimage analyses for use in digital twins

Siddhartha Nandy	<i>Spatio-temporal modeling of neuroimages using Covariate Assisted Principal Regression</i>
Ansu Chatterjee	<i>Rapid change detection in streaming neuroimage data</i>
Sanjay Chaudhuri	<i>An empirical likelihood-based method for functional neuroimaging</i>

4:30–6:30 PM, June 2 | Michigan League Ballroom, 2nd floor

Posters and Mixer Event

P2 Haozheng Xu	<i>A Curated Meta-analytic Neuroimaging Dataset from Neurovault</i>
P3 Chen Mu	<i>Statistical Analysis and Characterization of Gold Nanoparticle Morphology from Tomographic Image Stacks</i>
P4 Pascal Tyrrell	<i>Restoring Radiomic Feature Interpretability After Principal Component Analysis in Prostate MRI Cancer Risk Models</i>
P5 Khashayar Namdar	<i>AI-Powered Endometriosis MRI Reporting: Automated #Enzian Scoring in Pelvic MRI Enabled by Large Language Models</i>
P7 Susan Glenn	<i>Bootstrap-based Hypothesis Test of 2D Contours using Elastic Shape Analysis</i>
P8 Robert J Bishop	<i>JointSpace and Operating Rhythms</i>
P9 Stefan Eng	<i>Modeling Differences in Tumor-Immune Co-Localization to Predict Immune Checkpoint Inhibitor Response in Metastatic NSCLC</i>
P10 Xiaoyu Qiu	<i>Benchmarking Uncertainty Quantification of Plug-and-Play Diffusion Priors for Inverse Problems Solving</i>
P11 Martin Salgado-Flores	<i>Relational Persistent Homology of the Tumor Microenvironment for Survival Analysis of Lung Adenocarcinoma</i>
P12 Shashipraba Rajakaruna	<i>Topology guided conditional generative adversarial network for medical image simulation</i>
P13 Sanya Kejriwal	<i>Evaluating the Performance of the SPICE Test Under Heterogeneous Spatial Autocorrelation</i>

Posters and Mixer Event (continued)

P14 Jessica Aldous	<i>MoSAIC: Multi-Resolution Spatial Regression Analysis of Cellular Colocalizations in Cancer Imaging</i>
P15 Zihang Wang	<i>Flexible Normative Brain Modeling Using Quantile Super Learning with Motion Control</i>
P16 Yuting Duan	<i>Robust Image-on-scalar Regression with Simultaneous Activation Region and Extreme Signal Detection</i>
P17 Yuhan Geng	<i>Threshold Spatial Attention Transformer for Efficient Image Generation</i>
P18 Yiyan Hao	<i>Applying Multiple Imputation On Neuroimaging Data While Maintaining Spatial Autocorrelation</i>
P19 Rosa Fallahpour	<i>A Spatial Entropy Framework for Characterizing Radiomic Heterogeneity: Application to Healthy and Pediatric Tumor Brain MRI</i>
P20 John Bodenschatz	<i>Distributionally Accurate Phase Activation in fMRI</i>
P21 Ziyu Liu	<i>Bayesian Image-on-Image Regression for Linking Resting-State Connectivity to Task fMRI Activation</i>
P22 Longhao Pang	<i>Bayesian Joint Singular Value Decomposition with Application to Image Analysis</i>
P23 Yuxiao Nie	<i>Sparse Dynamic Latent Factor Models for Multi-Channel EEG based Brain Computer Interfaces</i>
P24 Grant Carr	<i>SPARCL: Spatially Adaptive Regression for Covariate Driven Clustering</i>
P25 Razmin Bari	<i>Extracting Bipolar-Associated Functional Connectivity Subgraphs in Adolescent rs-fMRI Using Adaptive Dense Subgraph Discovery</i>
P26 Siyan Wen	<i>Out-of-Sample CovBat for Multi-Site Harmonization of Resting-State Functional Connectivity</i>
P27 Mara Sherlin Talento	<i>KenCoh: A Rank-Based Estimator of Canonical Coherence</i>
P28 Frank Fazekas	<i>Centrosome Positioning and Nuclear Architecture: Implications for T Cell Activation</i>

4:30–6:30 PM, June 2 | Michigan League Ballroom, 2nd floor

Posters and Mixer Event (continued)

P29 | Minjin Lee

Monte Carlo–Based Assessment and Spatial Smoothing for CTRW Parameter Estimation in DWI

P30 | Ralph Jiang

Scalable Image-on-Scalar Regression for High-Dimensional Neuroimaging Data

Track A · S24 | 8:00–9:20 AM, June 3 | Michigan Room, 2nd floor

Network methods in neuroimaging

Tingting Zhang

Age-Related Changes in Structural Networks Across Large Neuroimaging Studies

Russel T. Shinohara

Multi-Site Harmonization Methods for Network-based Analyses

Daniel Kessler

Predicting Responses from Weighted Networks with Node Covariates in an Application to Neuroimaging

Alexander Huth

Mapping and Decoding Language Representations from Human Cortex

Track B · S25 | 8:00–9:20 AM, June 3 | Hussey Room, 2nd floor

AI for Medical Image Analysis III

Pascal Tyrrell

Restoring Radiomic Feature Interpretability After Principal Component Analysis in Prostate MRI Cancer Risk Models

Daljit Takher

Physics-Learning of Rotor Dynamics: A Reaction-Diffusion Video Prediction Framework for Cardiac Fibrillation

9:30–10:50 AM, June 3 | Michigan League Ballroom, 2nd floor

Student Paper Session: Theory & Methods

Zhenyao Ye (Winner)

Graph guided polygenic–pleiotropic architecture analysis for imaging-genetics data

Guoxuan Ma (Runner-up)

Sparse Bayesian modeling of EEG channel interactions improves P300 brain-computer interface performance

Student Paper Session: Case Studies & Applications

Yuzi Mi (Winner)

Bayesian functional spatial partitioning for boundary surface lesion detection using 3D MRI

Noah Hillman (Runner-up)

Testing for network specificity in brain-behavior associations using ordinal dominance curves

11:15 AM–12:15 PM, June 3 | Michigan League Ballroom, 2nd floor

Keynote Presentation #3

Douglas Noll

Spatiotemporal signal modeling in dynamic MRI



Scan the QR code to access
the Statistical Methods in
Imaging (SMI) 2026 website